

MIDDLE EAST TECHNICAL UNIVERSITY

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

EE201 – Circuit Theory I**Lecture 2: Kirchhoff's Laws**

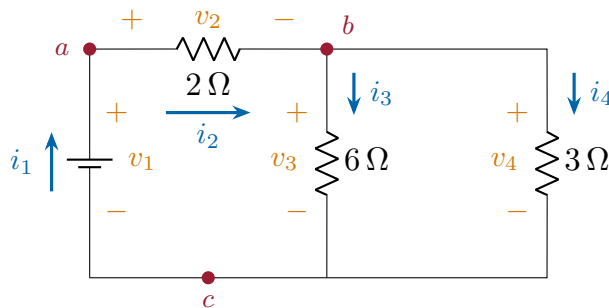
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Colour code used in these notes and in the slides

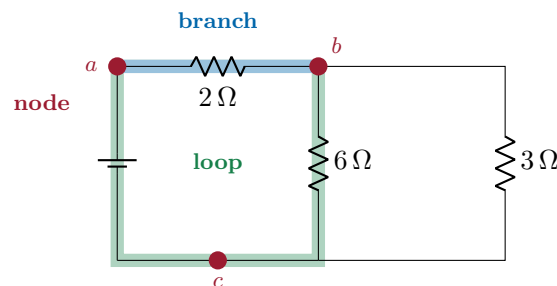
■ charge q
 ■ current i
 ■ voltage v
 ■ power p , energy w

1. A circuit, and the question it asks

Here is a circuit. Every element in it is one you already know, and the reference directions are fixed.

**1.1. Node, branch, loop**

Three words, all from Lecture 1, all about the *drawing* rather than the components.



Node	$n = 3$	a connection point. Everything joined by ideal wire is <i>one</i> node, no matter how much wire is drawn. Here: a, b, c .
Branch	$b = 4$	one element, joining two nodes. Here: the source and the three resistors.
Loop	—	a closed path that passes through no node twice. Here: the left-hand one, the right-hand one, and the outer one.

Throughout, n is the number of nodes and b the number of branches.

1.2. What we do not know

Each branch carries a voltage and a current, and none of the eight is known yet:

$$\underbrace{v_1, v_2, v_3, v_4}_{4 \text{ unknowns}} \quad \underbrace{i_1, i_2, i_3, i_4}_{4 \text{ unknowns}} \quad \Rightarrow \quad \boxed{2b = 8 \text{ unknown functions of } t}$$

1.3. Half the equations are easy

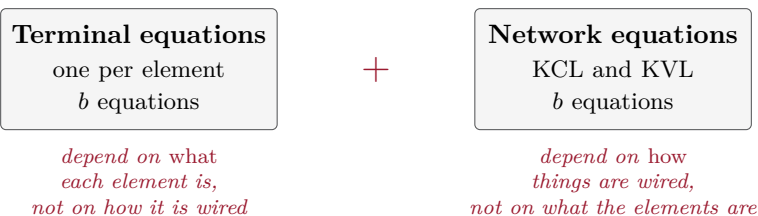
Each element brings one equation of its own, its **terminal equation**. It describes that element *in isolation* and knows nothing about the rest of the circuit:

$$v_1 = 12 \text{ V}, \quad v_2 = 2 i_2, \quad v_3 = 6 i_3, \quad v_4 = 3 i_4.$$

Element	Symbol	Terminal equation
resistor	R	$v = R i$
inductor	L	$v = L di/dt$
capacitor	C	$i = C dv/dt$
ind. voltage source	v_s	$v = v_s(t)$, i free
ind. current source	i_s	$i = i_s(t)$, v free

That is $b = 4$ equations for $2b = 8$ unknowns. **Exactly half.** Staring at the elements will not produce the other half: wire the same four components differently and every current changes, while not one terminal equation does.

1.4. The missing half comes from the wiring



The whole strategy of circuit theory

Cut the problem in two. The **elements** know nothing about the circuit they sit in; the **interconnection** knows nothing about the elements it joins. Kirchhoff's two laws are that second half.

Because they ignore the elements, KCL and KVL hold for *every* lumped circuit — linear or nonlinear, time-invariant or time-varying, at DC or at 100 MHz.

2. Kirchhoff's Current Law

KCL

For **every node** of a lumped circuit and at **every instant** t , the algebraic sum of the currents *leaving* the node is zero:

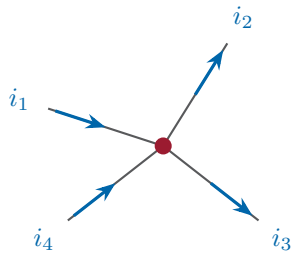
$$\sum_{k \in \text{node}} \pm i_k(t) = 0, \quad \text{sign } + \text{ if the arrow points away from the node.}$$

Equivalently: *what flows in, flows out.*

Why. A node is a junction of ideal wires; it has no volume and stores no charge. If $q_{\text{node}}(t) = 0$ for all t , then charge conservation gives

$$\frac{dq_{\text{node}}}{dt} = \sum_{\text{in}} i_k - \sum_{\text{out}} i_k = 0.$$

So KCL is conservation of charge plus the lumped assumption from Lecture 1 (*nothing piles up*).



Leaving positive:

$$-i_1 + i_2 + i_3 - i_4 = 0$$

Entering positive gives the same equation times -1 . Either convention works — use *one* of them all the way round.

Example 2.1 — KCL at a node

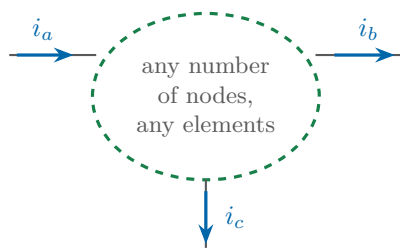
Four branches meet at a node with the reference arrows above. Measurements give $i_1 = 3$ A, $i_2 = -2$ A and $i_3 = 6$ A. Find i_4 .

$$-i_1 + i_2 + i_3 - i_4 = 0 \implies -3 + (-2) + 6 - i_4 = 0 \implies \boxed{i_4 = 1 \text{ A}}$$

Two lessons. The measured $i_2 = -2$ A enters *with its own sign*; we do not redraw the arrow. And the equation is written from the picture, not from any idea of which way the charge “really” goes.

2.1. KCL on a closed surface

Nothing in that argument needed the enclosed region to be a single point. Draw *any* closed surface: if the charge inside does not accumulate, the algebraic sum of the currents crossing it is zero.

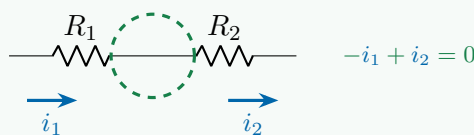


$$-i_a + i_b + i_c = 0$$

Whatever is inside — one node or a thousand, resistors or a whole amplifier — only the branches that *cross* the surface appear.

Example 2.2 — two consequences of the surface form

(a) **Series elements carry the same current.** Wrap a surface around the junction of two elements joined end to end. Exactly two branches cross it, so $i_1 = i_2$: a chain of elements has *one* current, and that is the definition of “in series”.



$$-i_1 + i_2 = 0$$

(b) **A two-terminal box returns everything it is given.** Put the surface around an whole sub-network reached by two wires: the current in at one terminal equals the current out at the other, however complicated the inside. That is what lets us speak of “*the* current through” a black box.

Two things KCL is not

- It is an equation about **one instant**. $\sum i_k(t) = 0$ holds for each t separately; nothing is averaged over time.
- It says nothing about voltages, resistances or power. Only topology and current.

3. Kirchhoff's Voltage Law

KVL

For **every loop** of a lumped circuit and at **every instant** t , the algebraic sum of the branch voltages around the loop is zero:

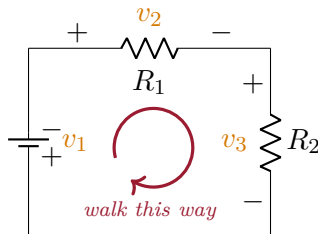
$$\sum_{k \in \text{loop}} \pm v_k(t) = 0, \quad \text{sign } + \text{ if you traverse the branch from } + \text{ to } -.$$

Equivalently: *go all the way round and you come back to where you started.*

Why. In a lumped circuit every node j can be given a single number e_j , its potential with respect to a chosen reference. A branch voltage is then a difference, $v_k = e_{(+)} - e_{(-)}$, and going round a loop $a \rightarrow b \rightarrow c \rightarrow a$ the sum telescopes:

$$(e_a - e_b) + (e_b - e_c) + (e_c - e_a) = 0.$$

Energetically: carry a charge once around the loop and it ends where it began, so the net energy exchanged per unit charge is zero. Like KCL, this is the lumped model at work: it holds as long as no magnetic field is changing through the loop, which is assumption (b) of Lecture 1.¹



$$-v_1 + v_2 + v_3 = 0$$

The source is traversed $-$ to $+$, so it enters with a minus sign; the two resistors are traversed $+$ to $-$ and enter with a plus. Walking the other way multiplies everything by -1 .

Voltage does not depend on the path

KVL is the same statement as: *the voltage between two nodes is the same whichever way you go between them.* If path 1 gives v_{ab} and path 2 gives v'_{ab} , then path 1 followed by path 2 backwards is a loop, so $v_{ab} - v'_{ab} = 0$. That is why “the voltage across R_2 ” means anything, and why node potentials exist at all.

Example 2.3 — one loop

$v_s = 12\text{ V}$, $R_1 = 2\Omega$, $R_2 = 4\Omega$ in the figure above. With the loop current i taken clockwise, the terminal equations give $v_2 = R_1 i$ and $v_3 = R_2 i$, and $v_1 = v_s$. KVL:

$$-12 + 2i + 4i = 0 \implies \boxed{i = 2\text{ A}}, \quad v_2 = 4\text{ V}, \quad v_3 = 8\text{ V}.$$

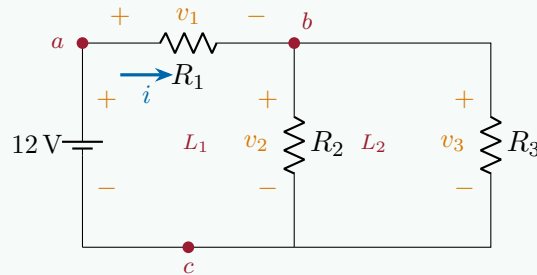
Check: $4 + 8 = 12\text{ V}$. The 12 V is shared between the two resistors, the larger one taking the larger share — a fact we will soon turn into a formula.

¹A transformer breaks that assumption on purpose: the loop picks up a voltage belonging to no element in the drawing. The cure is to draw the coupling as an element of its own, after which KVL holds exactly as written.

3.1. A loop that is not a mesh, and what “independent” means

Example 2.4 — three loops, two equations

Take the circuit of page 1 with $v_s = 12\text{ V}$, $R_1 = 2\ \Omega$, $R_2 = 6\ \Omega$, $R_3 = 3\ \Omega$.



Three loops can be drawn: L_1 , L_2 , and the outer one $L_3 = L_1 \cup L_2$. Walking each clockwise,

$$L_1 : -12 + v_1 + v_2 = 0, \quad L_2 : -v_2 + v_3 = 0, \quad L_3 : -12 + v_1 + v_3 = 0.$$

The third is the sum of the first two: *true*, but not *new*. Adding it to the system gains nothing.

Note what L_2 says on its own: $v_2 = v_3$. Two elements joined at *both* ends always carry the same voltage — they form a loop containing nothing else, and KVL closes it. (Section 5 solves this circuit completely.)

4. How many equations, and which ones?

Recall from Section 1.1: n is the number of **nodes**, b the number of **branches**. For the circuit of page 1, $n = 3$ and $b = 4$.

Source of equations	How many	Why that many
KCL, one per node	$n - 1$	the n -th is the sum of the others
KVL, one per loop	$b - n + 1$	$= \ell$, the independent loops
<i>network equations</i>	b	
terminal equations, one per element	b	
total	2b	$=$ number of unknowns v_k, i_k

Why is one KCL redundant? Each branch leaves one node and enters another, so its current appears in exactly two of the node equations — once with $+$ and once with $-$. Add all n equations together and every current cancels, leaving the empty statement $0 = 0$. That identity is what makes them dependent: whichever node you leave out, its equation is minus the sum of the other $n - 1$, so it can always be rebuilt from them and adds nothing. Only $n - 1$ carry information, and it does not matter which one you drop.

For the running example: $n = 3$, $b = 4$, so 2 independent KCL equations, 2 independent KVL equations, 4 terminal equations — 8 equations for the 8 unknowns $v_1..v_4, i_1..i_4$. We will soon turn this bookkeeping into systematic recipes — node and mesh analysis — that never write all $2b$ equations at once.

Some diagrams have no solution at all

KCL and KVL do not only solve circuits; they also decide which drawings *mean* anything.

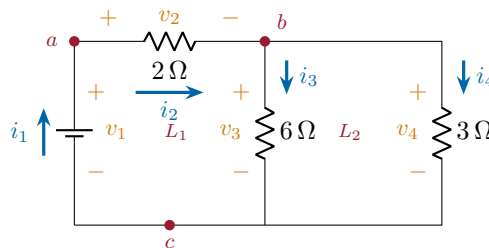
- Connect an ideal 5 V source directly across an ideal 9 V source. That loop contains nothing else, so KVL reads $5 - 9 = 0$. False — and no choice of currents can repair it.
- Put an ideal 2 A source in a chain with an ideal 3 A source. KCL at the node between them reads $2 = 3$. False again, whatever the voltages do.

Neither is an algebra mistake. The equations are contradictory, so the diagram describes no circuit at all. The *model* is what has gone wrong: an ideal source promises to hold its value whatever the current or voltage costs, and two such promises can disagree.

Real components never promise that much. Give each battery its internal resistance and the equations become consistent — and then tell you that a very large current circulates. Sometimes that is harmless; sometimes it is why one of the batteries makes smoke.

5. Solving the circuit of page 1

Collect everything. Four terminal equations, two KCL, two KVL — eight equations, eight unknowns, and no method beyond substitution.



4 terminal equations	2 KCL ($n - 1$)	2 KVL ($b - n + 1$)
$v_1 = 12$	node a: $-i_1 + i_2 = 0$	L_1 : $-v_1 + v_2 + v_3 = 0$
$v_2 = 2i_2$	node b: $-i_2 + i_3 + i_4 = 0$	L_2 : $-v_3 + v_4 = 0$
$v_3 = 6i_3$		
$v_4 = 3i_4$		

Eliminate in the order the equations suggest. L_2 gives $6i_3 = 3i_4$, so $i_4 = 2i_3$. KCL at b then gives $i_2 = i_3 + i_4 = 3i_3$. Substituting into L_1 ,

$$-12 + 2i_2 + 6i_3 = 0 \quad \text{with} \quad i_3 = \frac{1}{3}i_2 \quad \implies \quad -12 + 2i_2 + 2i_2 = 0.$$

Hence

$$i_1 = i_2 = 3 \text{ A}, \quad i_3 = 1 \text{ A}, \quad i_4 = 2 \text{ A}, \quad v_1 = 12 \text{ V}, \quad v_2 = 6 \text{ V}, \quad v_3 = v_4 = 6 \text{ V}.$$

A free check: power must balance

The source delivers $12 \times 3 = 36 \text{ W}$; the resistors absorb $2(3)^2 + 6(1)^2 + 3(2)^2 = 18 + 6 + 12 = 36 \text{ W}$. So $\sum_k p_k = 0$, as Lecture 1 promised. One line, and it catches most sign errors.

The eight-equation system always works — it is essentially what a circuit simulator solves — but it is slow by hand and grows fast: thirty branches would need sixty equations. We will soon learn much better ways to the same answer, writing only a handful of equations and satisfying KCL and KVL automatically.

Summary

Law	Statement	Rests on
KCL	$\sum \pm i_k = 0$ at every node, every instant	conservation of charge; nothing piles up
KVL	$\sum \pm v_k = 0$ round every loop, every instant	node potentials exist; no stray flux

- A circuit gives $2b$ unknowns and $2b$ equations: b **terminal** equations (what the elements are) and b **network** equations, $n - 1$ from KCL and $b - n + 1$ from KVL (how they are wired).
- KCL holds on **any closed surface**, not only at a node. Elements joined end to end carry one current; a two-terminal box returns what it takes.
- KVL is the statement that **voltage does not depend on the path**. Elements joined at both ends carry one voltage.
- Signs come from the *reference* arrows and polarities you drew, not from what the currents and voltages turn out to do. You may choose the references by guessing the true directions — a good guess keeps the answers positive — but nothing depends on it: the algebra uses the references as drawn, and a wrong guess simply returns a negative number.
- Neither law knows anything about the elements: both hold for nonlinear and time-varying circuits, exactly as written.

Next lecture

The other half of the equations: what a two-terminal element can be. Independent voltage and current sources, the general definition of a resistor, and the vocabulary for describing one — linear, bilateral, passive, controlled, time-invariant.

Reading

- Alexander & Sadiku, *Fundamentals of Electric Circuits*, Chapter 2.
- Chua, Desoer & Kuh, *Linear and Nonlinear Circuits*, Chapter 2.
- Lecture notes and videos of Prof. Emre Tuna, users.metu.edu.tr/etuna/ee201 — Lecture 2 video: network and terminal equations 6:50, KCL 8:35, KVL 18:55.